

Hossam Abu-elez Fathy

Machine Learning Engineer | Data Analyst | Data Scientist
Bachelor of Computer Science & Artificial Intelligence

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🐙 GitHub

🌐 LinkedIn

🌐 Portfolio

PROFILE

Computer Science and Artificial Intelligence student with hands-on experience in Machine Learning, Data Analysis, and Data Science. Experienced in building end-to-end AI solutions using Python, Scikit-Learn, SQL, and data visualization tools. Developed machine learning models achieving up to 99.46% accuracy on real-world datasets. Seeking internships and entry-level opportunities in Machine Learning, Data Analytics, and Data Science.

EDUCATION

•Artificial Intelligence Sciences

New Mansoura University

Oct 2023 – Jun 2027

CGPA: 3.4 / 4.0

EXPERIENCE

•Machine Learning Engineering Intern

FlyRank AI

Jul 2026 – Present

Remote

- Developing machine learning solutions for real-world AI applications.
- Building, evaluating, and improving ML models using Python.
- Collaborating with engineering teams on AI-driven projects.

•Data Analytics & BI Trainee

Information Technology Institute (ITI)

Jul 2026 – Present

Mansoura

- Building dashboards using Power BI and Tableau.
- Writing SQL queries and performing data analysis using Excel.
- Learning Data Warehousing and Business Intelligence concepts.

•Machine Learning Trainee

Creativa Innovation Hub

Jun 2025 – Aug 2025

Mansoura

- Preprocessed datasets and built ML models using Scikit-Learn.
- Applied feature engineering, classification, and regression techniques.
- Visualized data insights using Matplotlib and Plotly.

TECHNICAL SKILLS

Programming: Python, C++

Machine Learning: Supervised Learning, Unsupervised Learning, Deep Learning, Neural Networks, Machine Learning Algorithms, Model Evaluation, Model Selection, Feature Engineering, Data Preprocessing

Libraries: Scikit-Learn, Pandas, NumPy, Matplotlib, Plotly

Data Analysis: Excel, Power BI, EDA, Data Cleaning, Data Visualization

Databases: SQL, MySQL, PostgreSQL

Software Engineering: REST APIs, Git, GitHub, Object-Oriented Programming, Software Engineering Methodologies, Full-Stack Development

Tools: Git, GitHub, Markdown, Jupyter Notebook, VS Code, Kaggle

Soft Skills: Problem Solving, Communication, Team Collaboration, Analytical Thinking, Presentation, Adaptability, Continuous Learning

PROJECTS

- MedOptima AI Platform

AI Healthcare Decision Support System

– Developed an end-to-end AI healthcare decision support platform using Neural Networks, Knowledge-Based Systems, and Optimization Algorithms.

– Achieved **99.46%** test accuracy on a Kaggle dataset containing 41 diseases and 117 symptoms.

– Performed data preprocessing, feature engineering, hyperparameter tuning, and model evaluation.

– **Technologies:** Python, Scikit-Learn, SQL, REST APIs, Neural Networks, Full-Stack Development.
- Air Quality Prediction using Machine Learning

Machine Learning Regression Project

– Collected, cleaned, and preprocessed environmental datasets for predictive modeling.

– Built and evaluated regression models using Python and Scikit-Learn.

– Performed EDA and created interactive visualizations using Plotly and Matplotlib.

2024 – 2025
- AI Optimization Research Project - FTT Algorithm

Research and Optimization

– Designed and implemented the Football Team Training (FTT) Optimization Algorithm.

– Solved supply chain optimization problems and compared the proposed algorithm with existing methods.

– Improved convergence speed and solution quality.

2025
- Airline Reservation System

Full Software Engineering Project

– Designed system requirements, UML diagrams, and system architecture.

– Applied Object-Oriented Programming (OOP) principles and software engineering methodologies.

– Integrated a relational database for efficient data management.

2023 – 2024
- ALU 4-bit Digital Design Project

Computer Architecture Project

– Designed and implemented a 4-bit ALU supporting arithmetic and logical operations.

– Simulated and tested the ALU using digital logic design tools.

2024
- Computer Architecture & Assembly - 8085 & 8086

Assembly Programming

– Implemented and analyzed assembly programs using 8085 and 8086 instruction sets.

– Analyzed processor architecture, memory organization, and instruction execution.

2024

CERTIFICATES

- Machine Learning Training

Creativa Innovation Hub

Jun 2025
- Transact SQL Queries Using SQL Server

Mahara Tech

Mar 2025
- Communicate Effectively on GitHub using Markdown

Microsoft Learn

Apr 2026

LEADERSHIP & VOLUNTEERING

- AI & Programming Student Mentor

Supporting Students and Beginners

– Mentored students in AI, Python, and programming fundamentals.

– Guided beginners in problem-solving and technical skill development.

Present
- Tech Event Organizer

AI & Technology Workshops

– Organized AI and technology workshops for students.

– Coordinated event logistics and supported successful workshop execution.

Present

LANGUAGES

- Arabic: Native
- English: Proficient